Differentiation – Product Rule: Worked Examples

Product Rule

If $y = uv$ where *u* and *v* are functions of *x*, then

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Example

Differentiate:

$$y = 4x^2(e^{3x} + cosx)$$

$$u = 4x^2; \quad \frac{du}{dx} = 8x$$

$$v = e^{3x} + cosx; \quad \frac{dv}{dx} = 3e^{3x} - sinx$$

$$\frac{dy}{dx} = 4x^2(3e^{3x} - sinx) + (e^{3x} + cosx)(8x)$$

$$\frac{dy}{dx} = 12x^2e^{3x} - 4x^2sinx + 8xe^{3x} + 8xcosx$$

The result is often written in descending order

$$\frac{dy}{dx} = 12x^2e^{3x} + 8xe^{3x} - 4x^2sinx + 8xcosx$$

This can be factorised to

$$\frac{dy}{dx} = 4x(3xe^{3x} - xsinx + 2e^{3x} + 2cosx)$$